
National higher school of mathematics
SECOND MIDTERM EXAM

Algebra 1

2022 / 2023

1st Year Preparatory Cycle

Duration : 1h30min

The smartphone and calculator are forbidden.

Exercise 01

We define on $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$ the following two binary operations :

$$\forall (a, b), (c, d) \in \mathbb{Z}^2, (a, b) \oplus (c, d) = (a + c, b + d) \text{ and } (a, b) \otimes (c, d) = (ac, ad + bc).$$

1. Verify that the binary operation $\otimes$ is commutative.
2. Knowing that $(\mathbb{Z}^2, \oplus)$ is a commutative group with the neutral element $(0, 0)$, show that $(\mathbb{Z}^2, \oplus, \otimes)$ is a commutative ring with unity.
3. Determine the set D of zero divisors in $(\mathbb{Z}^2, \oplus, \otimes)$.
4. Determine the set U of invertible elements in $(\mathbb{Z}^2, \oplus, \otimes)$.
5. Let $B = \{(a, 0) : a \in \mathbb{Z}\}$. Show that B is a subring of $(\mathbb{Z}^2, \oplus, \otimes)$ without zero divisors.
6. Let f be the following map :

$$\begin{aligned} f : \mathbb{Z} &\longrightarrow \mathbb{Z}^2 \\ n &\longmapsto (n, 0). \end{aligned}$$

- (a) Show that f is a ring homomorphism from $(\mathbb{Z}, +, \cdot)$ to $(\mathbb{Z}^2, \oplus, \otimes)$.
- (b) Determine $\ker(f)$. What can we deduce?
- (c) Is f a ring isomorphism?

Exercise 02

We define on $\mathbb{R}^2$ the binary relation $\mathcal{R}$ as follows :

$$\forall (x, y), (x', y') \in \mathbb{R}^2, (x, y) \mathcal{R} (x', y') \iff \exists a > 0, \exists b > 0 \text{ such that } x' = ax \text{ and } y' = by.$$

1. Show that $\mathcal{R}$ is an equivalence relation.
2. We put $A = (1, 0)$, $B = (0, -1)$ and $C = (1, 1)$. Determine the equivalence classes $\overline{A}$, $\overline{B}$ and $\overline{C}$ and represent them geometrically.

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3. Determine all the equivalence classes and the cardinality of the quotient set $\mathbb{R}^2/\mathcal{R}$.
 4. Deduce the quotient set $\mathbb{R}^2/\mathcal{R}$.

Exercise 03

For each of the following statements, determine whether the statement is true or false and justify your answer.

1. For every two elements a and b of a group G we have $(ab)^{-1} = a^{-1}b^{-1}$.
2. Every cyclic group is commutative.
3. $n\mathbb{Z} + m\mathbb{Z} = \gcd(n, m)\mathbb{Z}$, where $n, m \in \mathbb{N}$.
4. $(\mathbb{C}^*, .)$ is isomorphic to $(\mathbb{R}^*, .)$.

Good luck